

ROMAIN BASTIANI

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Actively seeking a full-time position as a Quantitative Researcher.

Work Experience

Inception Technology

Quantitative Researcher - Macro and Systematic Commodities

Paris, France

July 2025 - April 2026

Start-up providing quantitative positioning intelligence tools to tier-1 global commodity trading houses (clients under NDA).

- Leveraged and enhanced a **macro factor lens** to deliver structural P&L attribution and systematic risk diagnostics for commodity derivatives portfolios.
- Developed a **regime-switching scenario engine** using **Gaussian Mixture Models (GMM)** validated through **Time-Series Cross-Validation** to capture non-linear factor dependencies and quantify tail positioning risks.
- Keywords:** Macro Factor Modeling, Regime Detection, Gaussian Mixture Models (GMM), Time-Series Cross-Validation, Tail Risk Diagnostics.

Quantitative Researcher Intern

Jan 2025 - July 2025

- Enhanced Inception's proprietary **CTA/Trend positioning model** by integrating a **Time Series Momentum (TSMOM)** framework inspired by **Moskowitz, Ooi and Pedersen (2012)** to model systematic exposure dynamics across commodity futures.
- Analyzed **CFTC COT data** (Disaggregated & Supplemental) to decompose Managed Money exposure into **CTA/Trend, Index Traders, and Discretionary** components.
- Developed positioning diagnostics including **capacity utilization, saturation zones, and flow convexity** to quantify asymmetric positioning risk and potential forced flows.
- Keywords:** COT Analysis, CTA Positioning, Time Series Momentum, Flow Convexity, Position Saturation.

Funds Avenue

Risk Management Intern

Luxembourg

July 2024 - Dec 2024

- Assessed portfolio risk using **stress tests** and **Value-at-Risk (VaR)** via Python/Excel models and presented risk reports to senior management.
- Applied multi-factor risk frameworks aligned with **Axioma** and **Barra**-style models for factor exposure analysis and systematic variance decomposition.
- Keywords:** Risk Management, Value-at-Risk (VaR), Stress Testing, Portfolio Analysis, Python, Axioma, Barra.

Extracurricular Projects

ENS / QRT Data Challenge - Asset Allocation Forecasting | Python, Machine Learning

Jun 2026 - Present

- Developed a structure-aware forecasting framework combining allocation-level models with target-free temporal constraints; achieved a **0.5472 public score** and currently rank **8th out of 1,426 participants**.

Gueant-Lehalle-Fernandez-Tapia (GLFT) Market Making Model | Python, NumPy, Pandas, Polars, SciPy

Sep 2025

- Implemented the closed-form optimal quoting strategy under inventory constraints from the Gueant-Lehalle-Fernandez-Tapia market making framework (extension of Avellaneda-Stoikov).
- Built a realistic backtest including Poisson fills, inventory dynamics, maker fees, latency and stochastic slippage.

Portfolio Optimization Strategy | Python, NumPy, SciPy, Matplotlib

Jan 2024

- Constructed optimal portfolios using Mean-Variance Optimization via `scipy.optimize` to maximize the Sharpe Ratio.
- Simulated thousands of portfolios (Monte Carlo) to visualize the Efficient Frontier and the Capital Market Line (CML).

Education

Neoma Business School

MSc Corporate Finance & Investment Banking | GPA: 4.0/4.0

Rouen, France

Oct 2023 - Dec 2024

- Double degree with ESIGELEC.
- Thesis: **Stochastic Models and Real Options in Corporate Valuation.**
- Key Modules:** Financial Engineering, Advanced Corporate Finance, Mergers & Acquisitions, Corporate Strategies.

ESIGELEC

MEng of Science, Applied Mathematics | Top 10% of Class

Rouen, France

Sep 2018 - Dec 2024

- Key Modules:** Advanced Probability, Machine Learning, Monte Carlo Simulation, Applied Statistics for Finance.

Additional Training

MIT - edX

Quantitative Finance & Machine Learning Certifications

Online

- Mathematical Methods for Quantitative Finance:** Probability, stochastic processes (discrete & continuous), Itô calculus, time series.
- Machine Learning with Python:** Linear models, clustering, recommender systems, CNN, generative models, reinforcement learning, NLP.

Skills & Interests

Languages: French (native), English (C1)

Programming: Python (NumPy, pandas, scipy, scikit-learn, Polars), SQL

IT: Microsoft Office, L^AT_EX

Interests: Fencing (Regional Level & Referee), French History